# Combinatorics Cameron - Solutions Manual

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# Contents

# Chapter 1

# On Numbers and Counting

### Exercise 1

Missing proof of existence of largest natural number, so cannot say "let $n$ be the largest...".

### Exercise 2

Fuh naw.

### Exercise 3

(a) Clearly, the base case works: $1 > \frac{1}{e}$. Now suppose

$$n! > \left(\frac{n}{e}\right)^n$$

Then

$$(n+1)n! > (n+1)\left(\frac{n}{e}\right)^n$$

We derive the right hand side:

$$= (n+1)\left(\frac{n}{n+1}\right)^n \left(\frac{n+1}{e}\right)^n$$

Using the fact that $(1+\frac{1}{n})^n < e \quad \Rightarrow \quad \frac{1}{(1+\frac{1}{n})^n} > \frac{1}{e} \quad \Rightarrow \quad \left(\frac{n}{n+1}\right)^n > \frac{1}{e}$, we replace the middle term to create the inequality

$$> (n+1)\frac{1}{e}\left(\frac{n+1}{e}\right)^n$$

$$= \left(\frac{n+1}{e}\right)^{n+1}$$

Hence induction step completed.
(b) The first inequality is equivalent to

$$\sqrt[n]{n!} < \frac{n+1}{2}$$

The left side is the geometric mean from 1 to $n$. The right side is the arithmetic mean from 1 to $n$. The theorem tells us that the geometric mean is smaller than the arithmetic mean.

Once again, we use the same strategy as (a).

$$\begin{aligned} n! &< \left(\frac{n+1}{2}\right)^n \\ &= \left(\frac{n}{n}\right)^n \left(\frac{n+1}{2}\right)^n \\ &= \left(\frac{n+1}{n}\right)^n \left(\frac{n}{2}\right)^n \\ &= \left(1+\frac{1}{n}\right)^n \left(\frac{n}{2}\right)^n \\ &< e\left(\frac{n}{2}\right)^n \end{aligned}$$

Hence proven.

## Exercise 4

(a) Given, $\log x$ and $x^c$ for $c > 0$, since both go to infinity, we apply L'Hopital's

$$\lim \frac{\log x}{x^c} = \lim \frac{\frac{1}{x}}{c\frac{1}{x^{1-c}}} = \frac{1}{c} \lim x^{-c} = 0$$

(b) Given $c, d > 1$, we know

$$\lim \frac{c^x}{x^d} = \infty$$

which is equivalent to the existence of $x$ such that $\frac{c^x}{x^d} > M$ for any $M$. Simply choose $M = 1$. Since $c^x$ is positive, in order to produce positive result greater than 1, it follows that $x^d$ must be positive. Hence, there exists $x$ such that $\frac{c^x}{x^d} > 1 \Rightarrow c^x > x^d$ with no sign flip.

## Exercise 5

Not sure what unlabelled means.

## Exercise 8

There are $2^n$ configurations of inputs. For each of the $2^n$ configurations, we have 2 output options, either true or false, hence $2^{2^n}$ boolean functions.

## Exercise 9

I think studying set theory would be more beneficial and effective than this question. Skipped.

## Exercise 10

$$\frac{g(n)}{f(n)} > \frac{kc^n}{bn^d}$$

for some $b, k$ and sufficiently large $n$. By L'Hopital's, the right limit tends to infinity, hence there exists $N$ such that the right side is greater than 1 for all $n \geq N$ (similar to Exercise 4). Hence $g(n) > f(n)$ for $n \geq N$.

## Exercise 11

For the first, since $b$ subsets each with $k$ elements, then the combined total count of elements of $bk$. For the second, there are $v$ elements, each found in exactly $r$ subsets, which adds up to $vr$, which is also the combined total count of elements in subsets.

## Exercise 12

(i) Too lazy. Pretty intuitive and obvious.

## Exercise 13

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## Exercise 14

Number of books:

$$B = 25^{410 \cdot 40 \cdot 80}$$

Number of rooms:

$$R = \frac{B}{20 \cdot 35}$$